

Output Form (OF)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: SAMSUM | Context: South Asia and MENA — Clinical NLP Researchers and AI Practitioners

Output Form definition: Whether the expected output modality matches regional deployment needs and accessibility.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Primary evaluation uses ROUGE F1 (R-1, R-2, R-L) [Q47] supplemented by a -1/0/1 human rating scale [Q48]. The benchmark itself documents that ROUGE is misleading for dialogue summarization — high ROUGE corresponds with low human ratings [Q3, Q54, Q67, Q68] — and explicitly calls for new metrics [Q69, Q74] without providing them. For SA-MENA deployment, this dialogue-summarization unreliability is compounded by complete inapplicability to non-English outputs: ROUGE/BERTScore against English references provide no valid signal for Arabic, Hindi, Urdu, Farsi, or Dari per the regional YAML and user elicitation. WEB-13 (AraHealthQA 2025) directly corroborates that BERTScore fails to capture clinical appropriateness for Arabic mental health tasks. Empirical sampling confirms 100% English content (DATASET-D19, DATASET-D17), with no multilingual signal whatsoever. HIGH priority weight on OF and complete cross-lingual misalignment yield a severe external validity failure.

ID	Evidence
Q3	"We show that model-generated summaries of dialogues achieve higher ROUGE scores than the model-generated summaries of news – in contrast with human evaluators' judgement." (p.1)
Q47	"We evaluate models with the standard ROUGE metric (Lin, 2004), reporting the F1 scores (with stemming) for ROUGE-1, ROUGE-2 and ROUGE-L following previous works (Chen and Bansal, 2018; See et al., 2017)." (p.4)
Q48	"We asked two linguists to mark the quality of every summary on the scale of -1, 0, 1, where -1 means that a summarization is poor, extracts irrelevant information or does not make sense at all, 1 – it is understandable and gives a brief overview of the text, and 0 stands for a summarization that extracts only a part of relevant information, or makes some mistakes in the produced summary." (p.5)
Q54	"Models generating dialogue summaries can obtain high ROUGE results, but their outputs are marked as poor by human annotators." (p.5)
Q67	"We show that the most popular summarization metric ROUGE does not reflect the quality of a summary." (p.8)
Q68	"We performed an independent, manual analysis of summaries and we demonstrated that high ROUGE results, obtained for automatically-generated dialogue summaries, correspond with lower evaluation marks given by human annotators." (p.8)
Q69	"We conclude that when measuring the quality of model-generated summaries, the ROUGE metrics are more indicative for news than for dialogues, and a new metric should be designed to measure the quality of abstractive dialogue summaries." (p.8)
Q74	"In order to perform well, it may require designing dedicated tools, but also new, non-standard measures to capture the quality of abstractive dialogue summaries in a relevant way." (p.9)
D17	Eva: Meeting today with clients at 11 am sharp.. / Jim: Yes sir / Jim: Yes sir I have — Semi-formal workplace exchange; conventional English register
D19	Tom: Did anyone pick up prints? / Glen: I didn't have time, but called John? — Minimal workplace coordination; turn-by-turn structure with speaker attribution
WEB-13	AraHealthQA 2025 found BERTScore 'failed to fully measure appropriateness or trustworthiness' for Arabic mental health QA — direct corroboration that automated metrics are insufficient (https://arxiv.org/html/2508.20047v2)

Strengths

- The benchmark explicitly documents ROUGE's unreliability for dialogue summarization [Q3, Q67, Q68], providing authored justification for users to supplement or replace the metric

- Manual human rating procedure with conflict resolution [Q48, Q49] establishes a precedent for human-in-the-loop evaluation that aligns with WEB-13's recommendations for Arabic mental health

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Q49	"We noticed a few annotations (7 for news and 4 for dialogues) with opposite marks (i.e. one annotator judgement was –1, whereas the second one was 1) and decided to have them annotated once again by another annotator who had to resolve conflicts." (p.5)
Q67	"We show that the most popular summarization metric ROUGE does not reflect the quality of a summary." (p.8)
Q68	"We performed an independent, manual analysis of summaries and we demonstrated that high ROUGE results, obtained for automatically-generated dialogue summaries, correspond with lower evaluation marks given by human annotators." (p.8)
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Checklist

Checklist item definitions:

ID	Assessment Question
OF-1	Does output modality match regional deployment needs?
OF-2	Assess text-to-speech availability for speech output
OF-3	Consider literacy rates and accessibility requirements
OF-4	Document form mismatches harming external validity

Checklist responses:

OF-1: Expected output modality is English-language plain text scored by ROUGE against single English reference. Regional deployment requires Arabic, Hindi, Urdu, Farsi, Dari, Bengali, Pashto outputs — none supported.

OF-2: Not directly applicable (text-to-text task), but downstream clinical deployment may require speech inputs/outputs; the benchmark provides no signal here.

OF-3: Literacy and accessibility considerations: regional context notes ~1 in 4 women in Pakistan/Bangladesh cite literacy as a mobile-connectivity barrier (WEB-2); the benchmark's text-only English form does not engage with this constraint.

OF-4: External validity is severely harmed: ROUGE is unreliable even for the English dialogue task by the authors' own admission [Q67], and provides zero signal for non-English clinical outputs. Users would need entirely separate multilingual evaluation infrastructure (regional YAML, user elicitation), which the benchmark does not provide.

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Q67	"We show that the most popular summarization metric ROUGE does not reflect the quality of a summary." (p.8)
Q69	"We conclude that when measuring the quality of model-generated summaries, the ROUGE metrics are more indicative for news than for dialogues, and a new metric should be designed to measure the quality of abstractive dialogue summaries." (p.8)
WEB-2	Literacy barriers affecting mental health access in Pakistan/Bangladesh — accessibility considerations not addressed by benchmark (https://datareportal.com/reports/digital-2024-deep-dive-the-state-of-internet-adoption)
WEB-13	AraHealthQA 2025 found BERTScore 'failed to fully measure appropriateness or trustworthiness' for Arabic mental health QA — direct corroboration that automated metrics are insufficient (https://arxiv.org/html/2508.20047v2)

WEB-15	MentalCLOUDS supplements ROUGE/BERTScore with expert evaluation across six clinical parameters — shows the form of evaluation SAMSum lacks (https://mental.jmir.org/2024/1/e57306)
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Information Gaps

- Whether any cross-lingual BERTScore variant has been validated specifically for clinical dialogue summarization in any target language (regional YAML notes this is unestablished)

Requires Expert Verification

- Which specific human evaluation rubric dimensions SA-MENA practitioners would prioritize when supplementing ROUGE

Remediation

Gap 1: ROUGE/BERTScore against English references provide no valid signal for non-English outputs and are documented as unreliable even for English dialogues by the authors themselves.

Recommendation: Replace ROUGE-only evaluation with a multi-tier framework: (i) target-language reference corpora with cross-lingual metrics (e.g., AraBERT-based BERTScore, IndicBERT) where validated; (ii) human-in-the-loop expert clinical evaluation per WEB-13 and WEB-15 across rubric dimensions including clinical appropriateness, cultural fit, risk preservation, and stigma-sensitive framing; (iii) error-typology audits for safety-critical content (suicide, abuse, self-harm) given DATASET-D6, DATASET-D10 demonstrate concrete failure modes.

ID	Evidence
D6	Pearl's boyfriend mistreats her but she loves him anyway. Patrick is trying to convince her to leave him. Pearl is fat and ugly, so it's not so easy for her. — Summary reproduces a derogatory self-description as factual without clinical reframing — illustrates embedded Western annotator salience norms
D10	Mickey tries to calm her with the fact that more people commit suicide on the Golden Gate Bridge than in that forest. — Suicide statistics treated as social reassurance in summary — no clinical risk annotation or flagging
WEB-13	AraHealthQA 2025 found BERTScore 'failed to fully measure appropriateness or trustworthiness' for Arabic mental health QA — direct corroboration that automated metrics are insufficient (https://arxiv.org/html/2508.20047v2)
WEB-15	MentalCLOUDS supplements ROUGE/BERTScore with expert evaluation across six clinical parameters — shows the form of evaluation SAMSum lacks (https://mental.jmir.org/2024/1/e57306)